

Day 4 GPUs and Cooling and Systems Theory

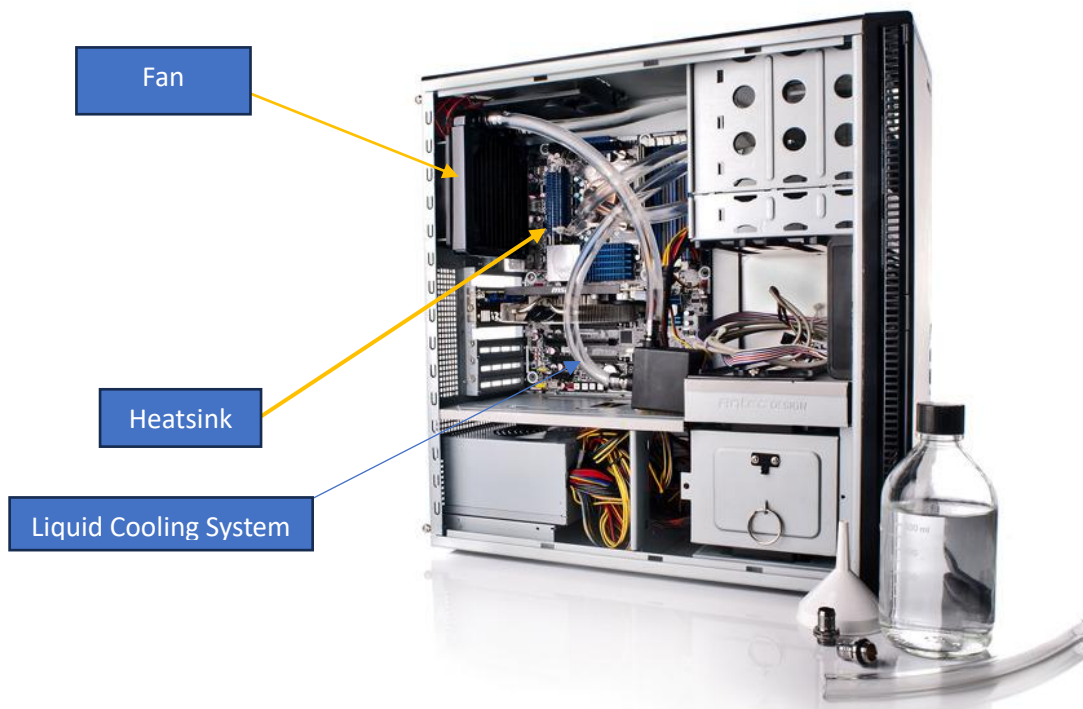
1. Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

GPU Details

- **Brand:** NVIDIA
- **Model:** GeForce RTX 3050 Ti
- **VRAM:** 4 GB GDDR6
- **Type:** Dedicated GPU
- **Use:** Gaming, graphics rendering, video editing, and other GPU-intensive tasks.



2. Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.



Cooling System Description

- **Fan:** The fan creates airflow inside the cabinet. It brings cool air into the case and pushes hot air out, helping to maintain a suitable internal temperature.
 - **Heatsink:** The heatsink is usually made of metal and is placed over or near the CPU. It absorbs heat from the processor and spreads it through its fins, where the airflow from the fan can carry the heat away.
 - **Liquid Cooling System:** Liquid cooling uses a coolant that circulates through tubes between the CPU block and radiator. The coolant absorbs heat from the CPU and carries it to the radiator, where the heat is released into the surrounding air.
3. Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Factor	Air Cooling	Liquid Cooling
cost	Cheaper	More expensive

efficiency	Good cooling but when heavy loads it's less Efficient.	Provide better cooling , Especially for high-performance.
maintenance	Low Maintance	High Maintance needs Check for liquid leaks, Pump failure
Noise	Fan become nosiy when CPU is in high loads	Can be quite because heat cool by liquid.
Installation	Easy To install	Complex to install

4. Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

I choose Liquid Cooling. Because This Tasks like streaming Ipl match and playing high graphics game generally generates heat very much and so we need more cooling to cool our cpu and gpu. So it can work on high or performance.

At High Temp of Gpu there are always decrease it's performance to maintain gpu parts otherwise overheat can also damage the gpu parts. And If gpu performance gonna decrease then Ipl steram may be buffer and games fps gonna drop in major. User experience gonna be worse.

So from prevent all this we use liquid cooling for our gpu over then air cooling. And also this is slient cooling system.